Reasoning LLMs

From First Principles to a From-Scratch R1 Recipe

Lucas Soares · O'Reilly Live Training

Agenda

1. **Intro to Reasoning LLMs** — what they are, why now
2. **The DeepSeek R1 Paper** — reading the recipe
3. **The R1 Recipe in Code** — five stages, three notebooks

01

Intro to Reasoning LLMs

What they are, why now

What do we mean by "reasoning"?

Recall — look something up:

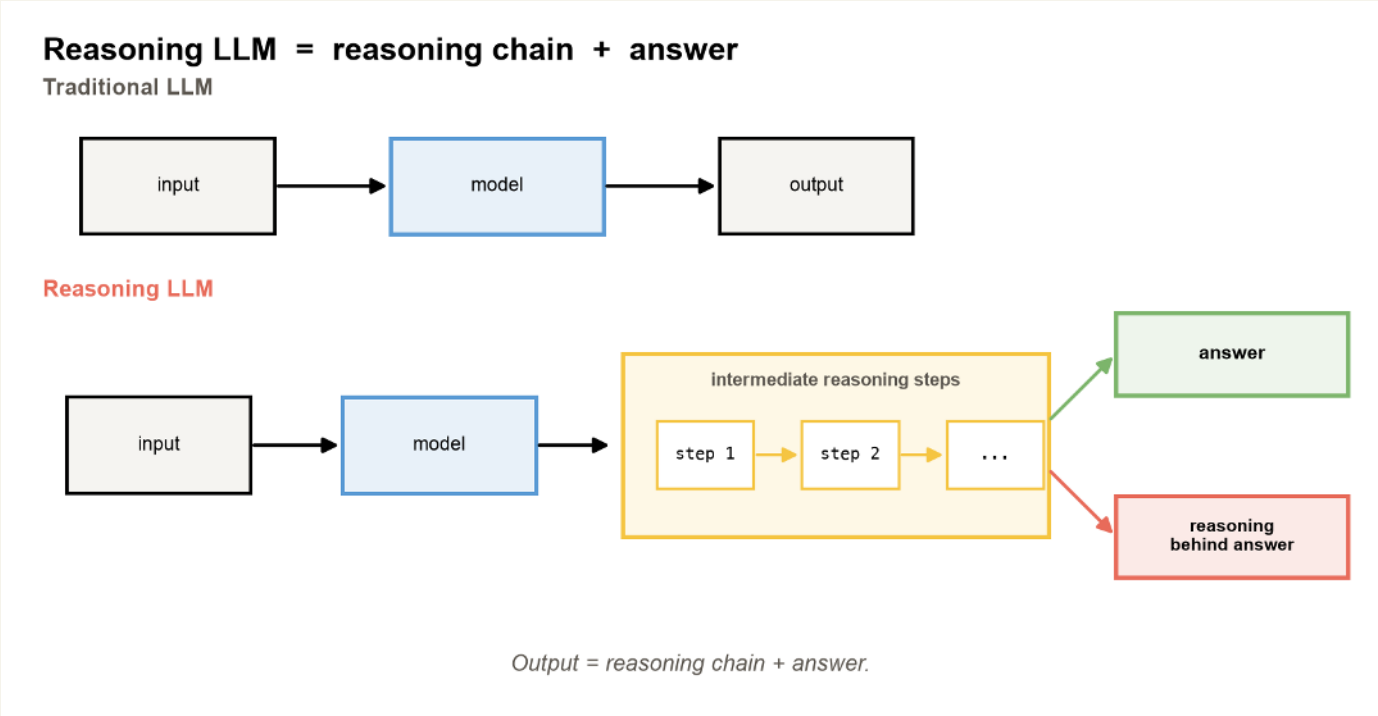
> "What's the course code for Stanford's Transformers class?" → CME 295

Reasoning — work it out:

> "A bear was born in 2020. How old in 2025?" → 2025 − 2020 = **5**

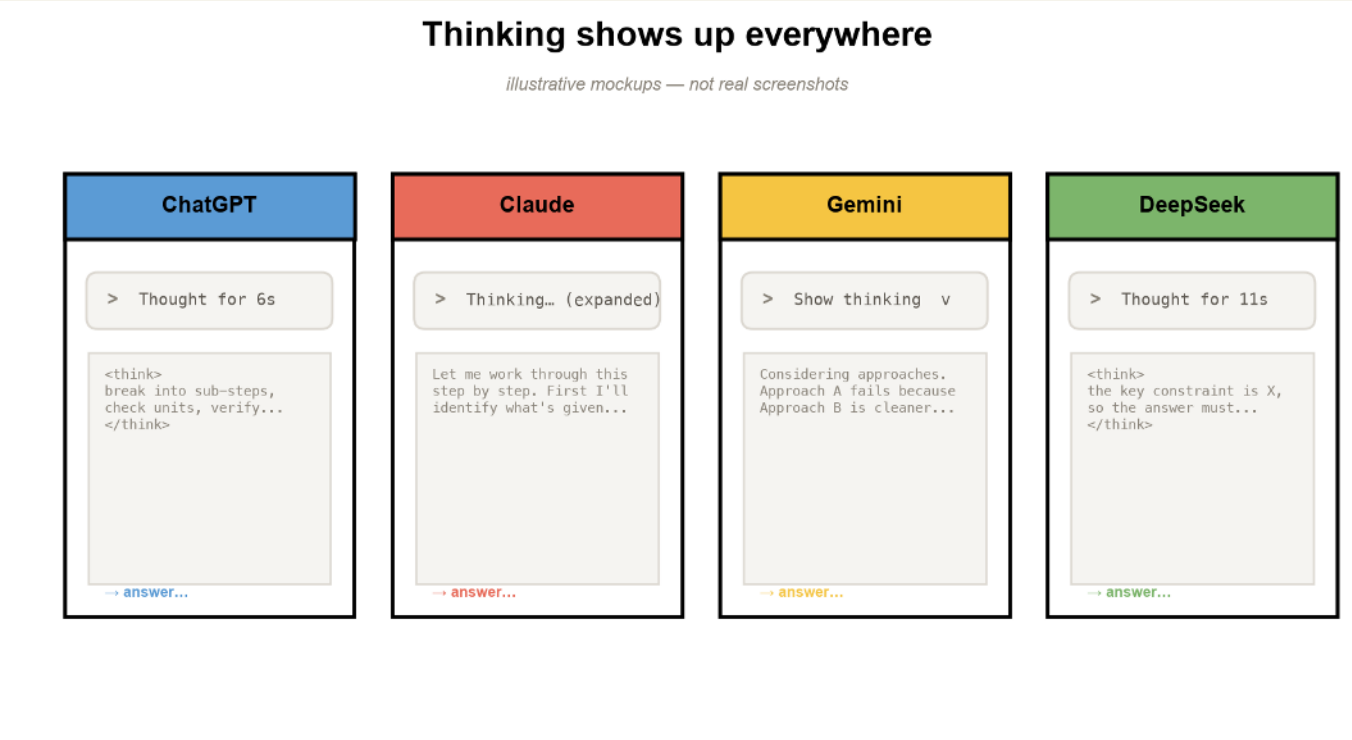
Reasoning = solving a problem through a **multi-step process**, not a single lookup.

What is a reasoning LLM?



The output contract changes: **Output = reasoning chain + answer.**

You've already seen these everywhere



The **"Thinking..."** tag is the tell — ChatGPT, Claude, Gemini, DeepSeek all show it.

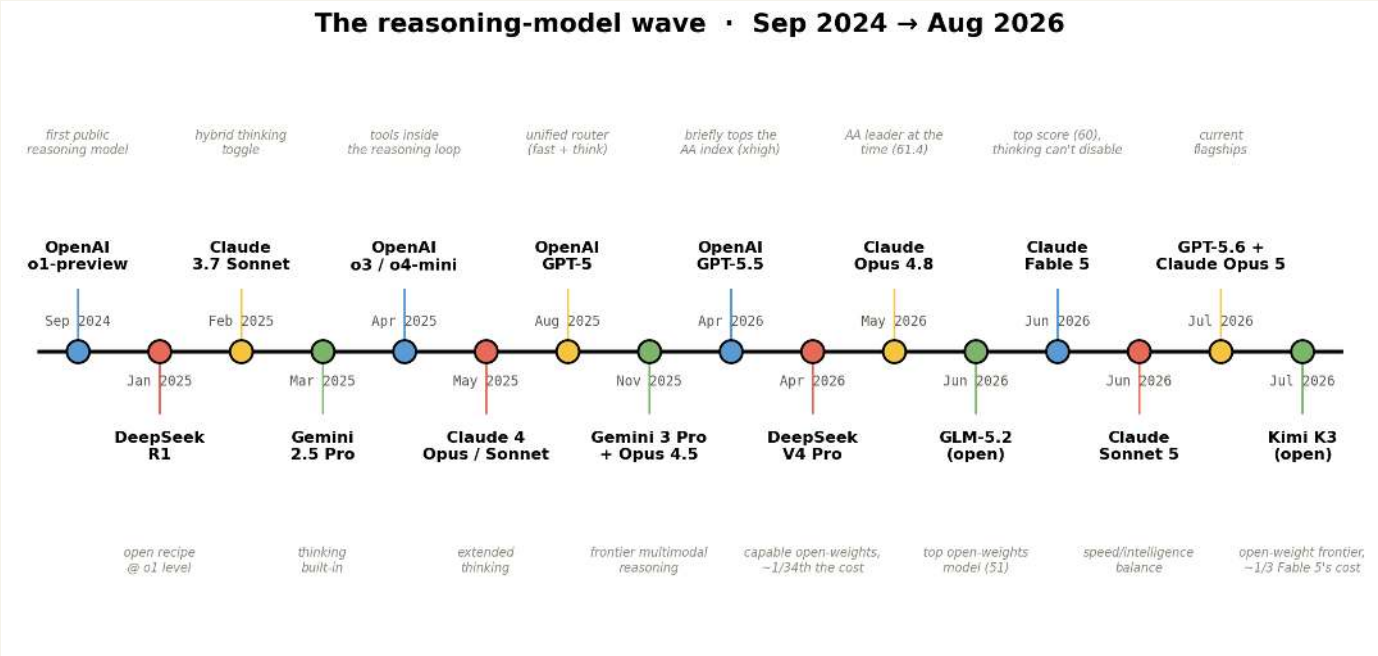
A catch: you pay for the thoughts

Reasoning tokens are **billed as output tokens** — on every major API.

- UIs show a *summarized* thought, not the raw chain (barely-readable, and exposed chains could train rivals)
- So there's a real incentive to get **the most reasoning per token**

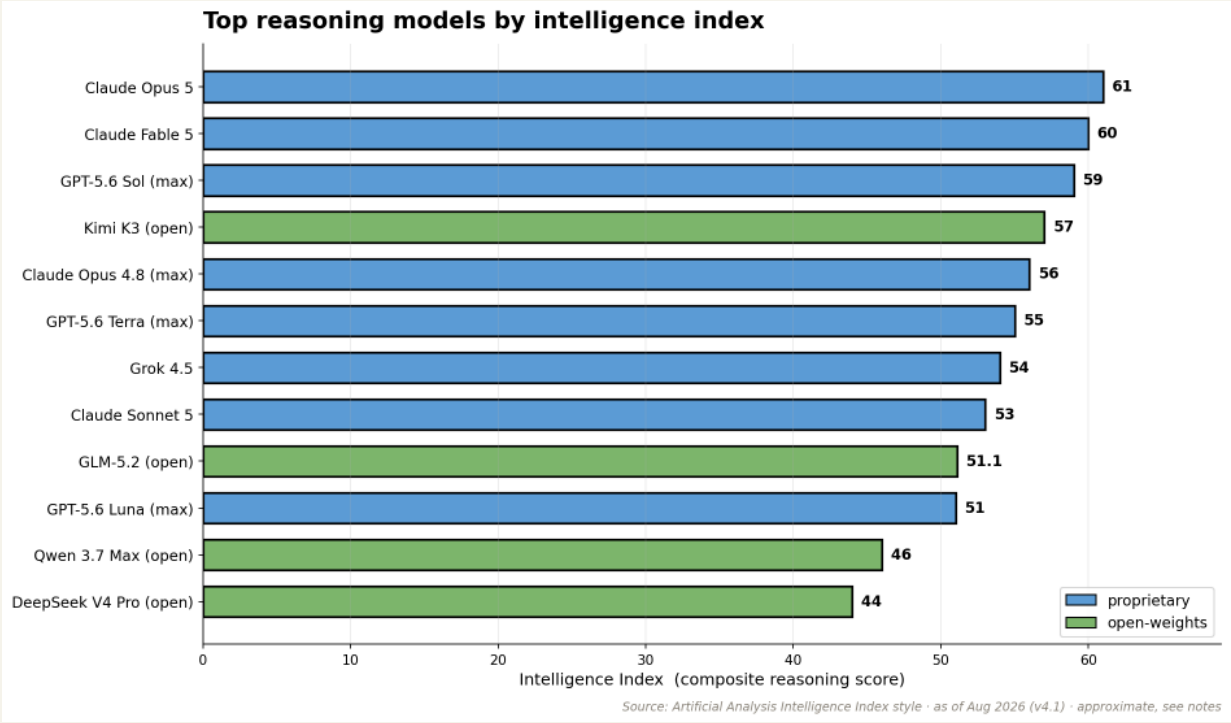
> More on controlling the thinking budget later.

How we got here



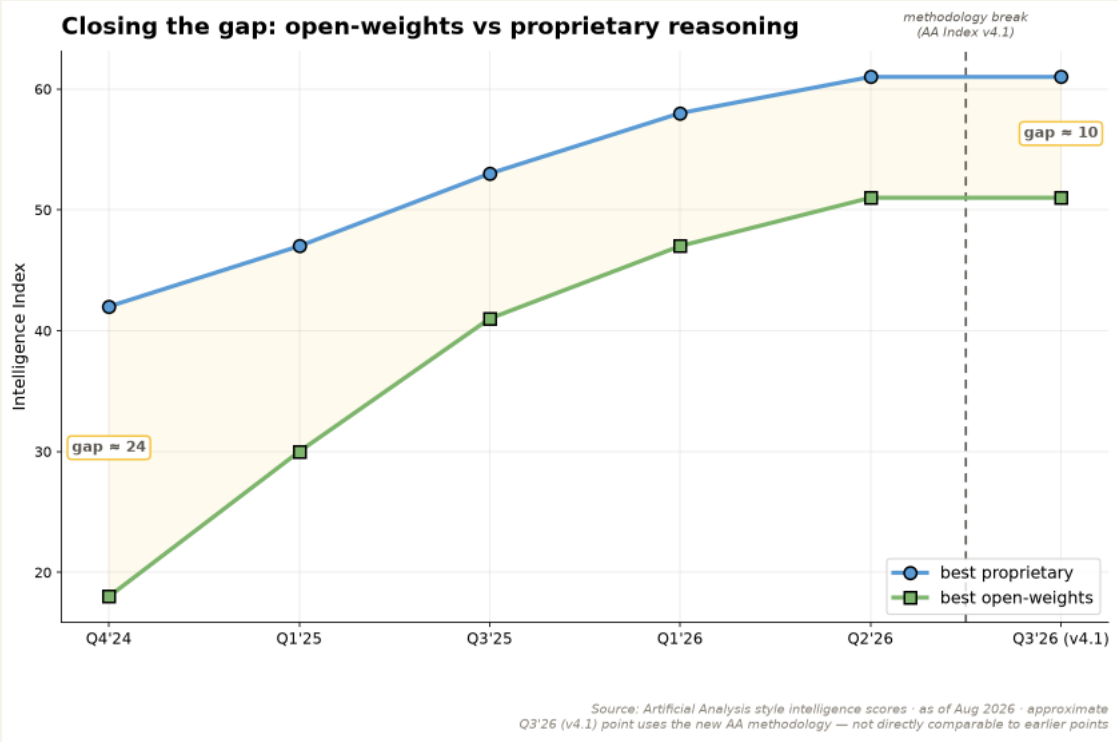
o1-preview kicked it off (Sep 2024). DeepSeek-R1 (Jan 2025) made the **recipe public**.

Where the models rank today



Demand concentrates on a handful of top labs. *(Artificial Analysis, ~Aug 2026, Intelligence Index v4.1.)*

Closing the gap



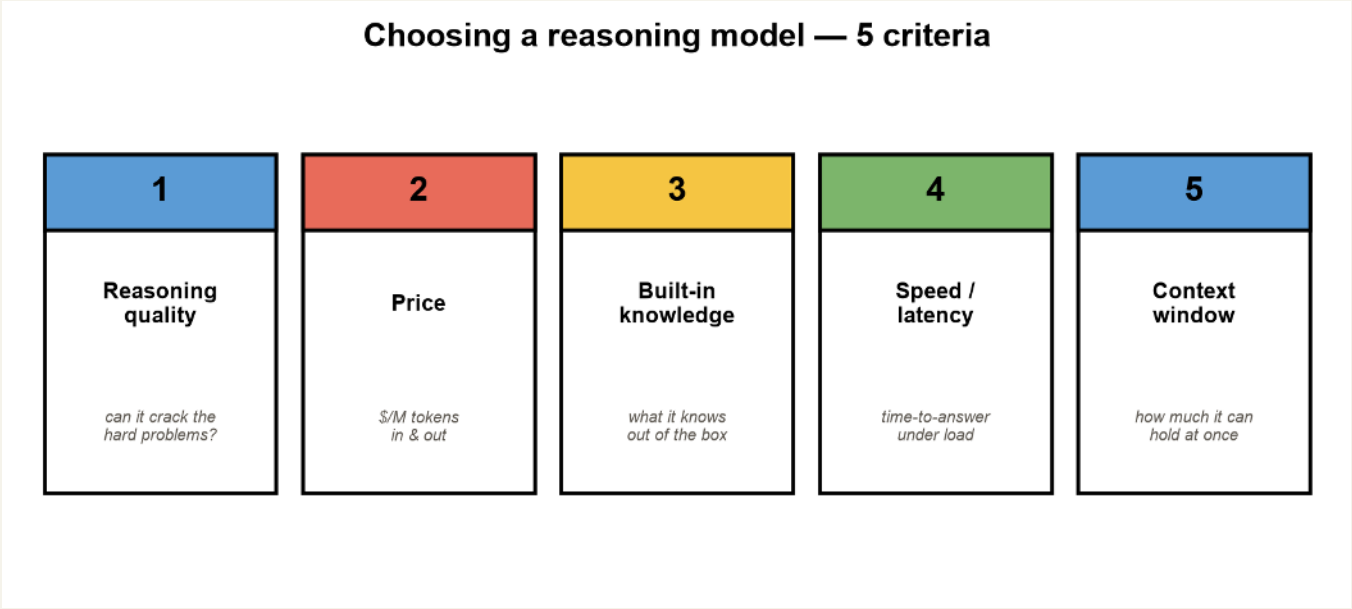
Open-weights reasoning models now trail the frontier by **months, not years**.

Why reach for a reasoning model?

It earns its extra tokens when the task is a **derivation**, not a lookup:

- Decomposes hard, multi-constraint problems
- Hallucinates less on problems it can *check itself* on
- Gives you the **reasoning**, not just the verdict

What actually matters when you pick one



Reasoning quality · price · built-in knowledge · speed · context window.

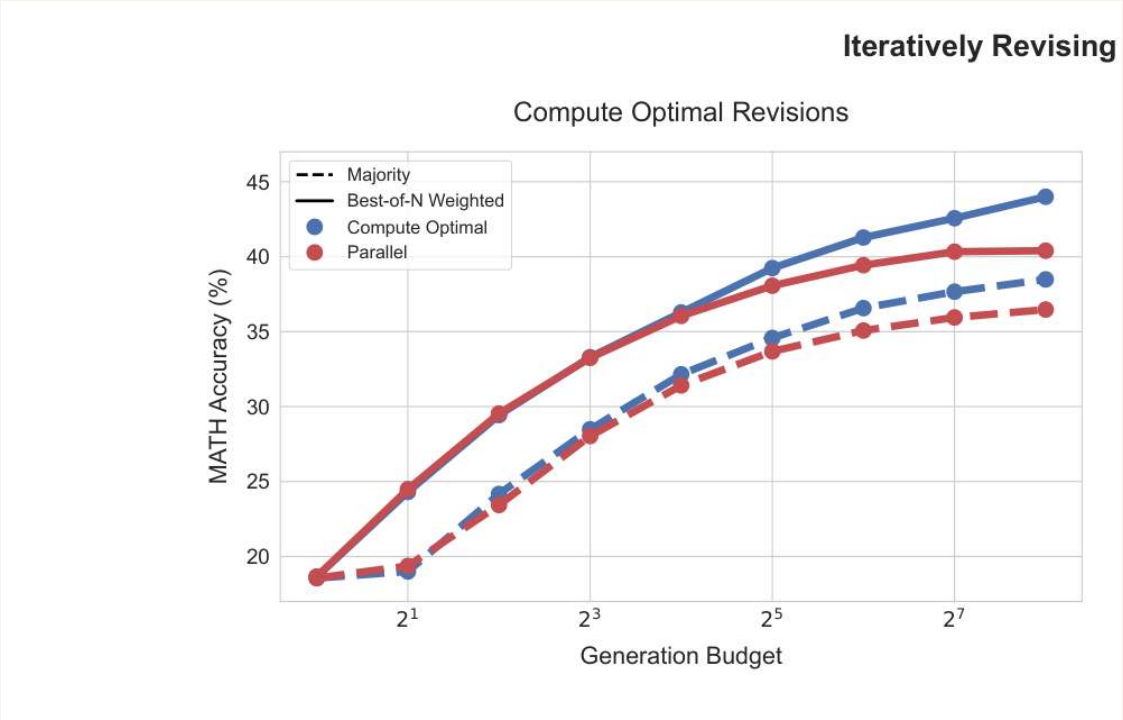
Price is the other axis

Claude Fable 5 **~$50** priciest, no off-switch	**GPT-5.6 Sol** **~$30** frontier	**Claude Opus 5** **~$25** frontier
Kimi K3 **~$15** open-weight, ~1/3 Fable 5's cost	**Gemini 3.1 Pro** **~$12**	**DeepSeek V4 Pro** **~$0.87** capable open-weights, ~34× cheaper

The expensive model isn't always the right call. *(Approx; we test this in notebook 05.)*

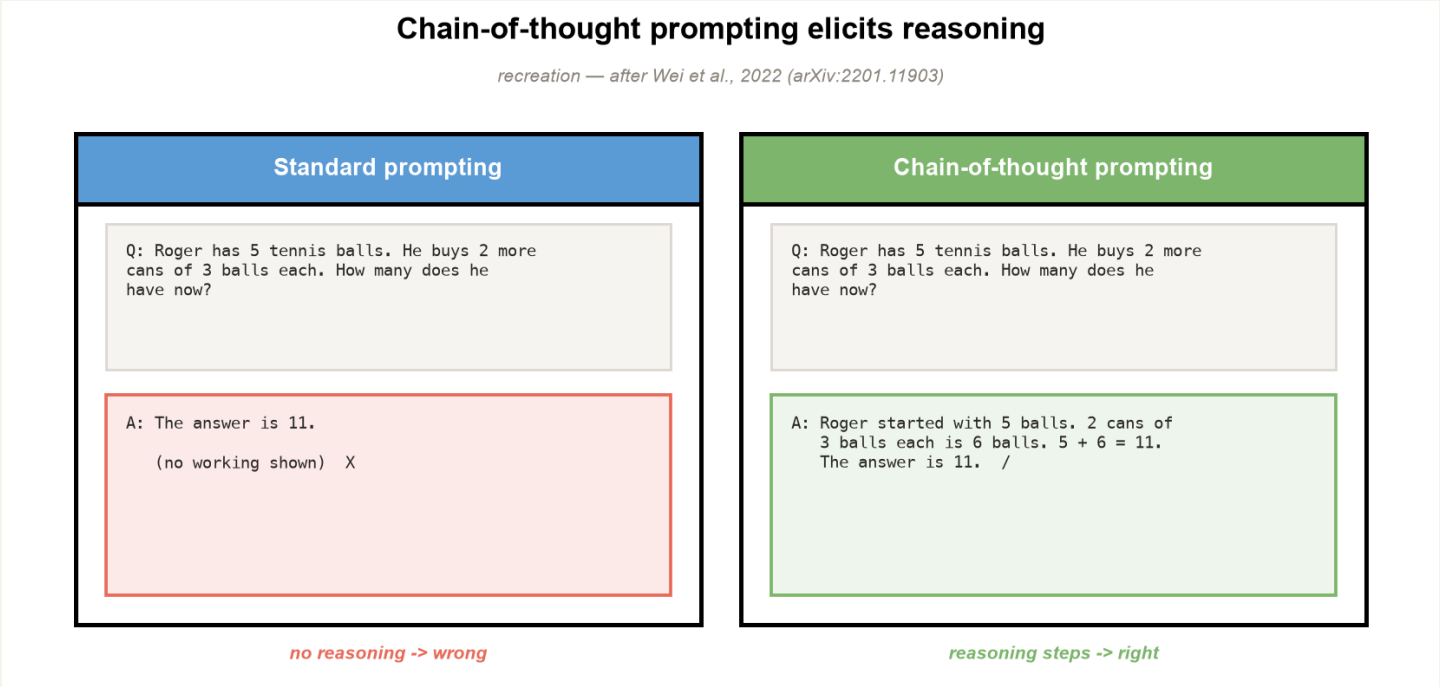
These prices will drift — for live numbers: artificialanalysis.ai/models/recommend

Why does thinking longer help?



Every generated token is another forward pass — reasoning tokens **buy compute** before the model commits.

The simplest version of the trick



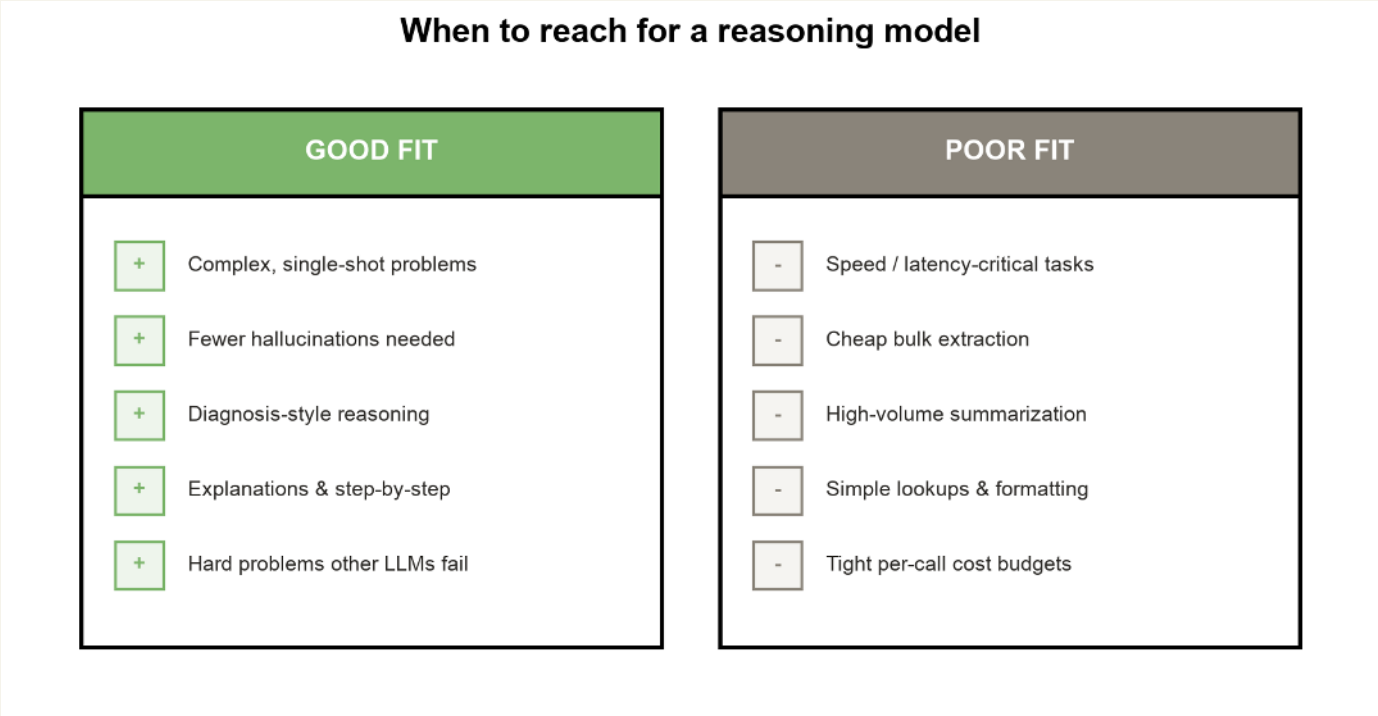
"Let's think step by step." Chain-of-thought, scaled up, *is* the core idea. *(after Wei et al., 2022)*

Two reasons a reasoning chain helps

1. **Decomposition** — break a problem the model has never seen into sub-problems it *has*.
2. **More compute** — more tokens before the answer = more computation spent on it.

Extended chains and self-correction ("wait, that's wrong…") **emerge from RL**, not just from prompting.

When to use one — and when not to

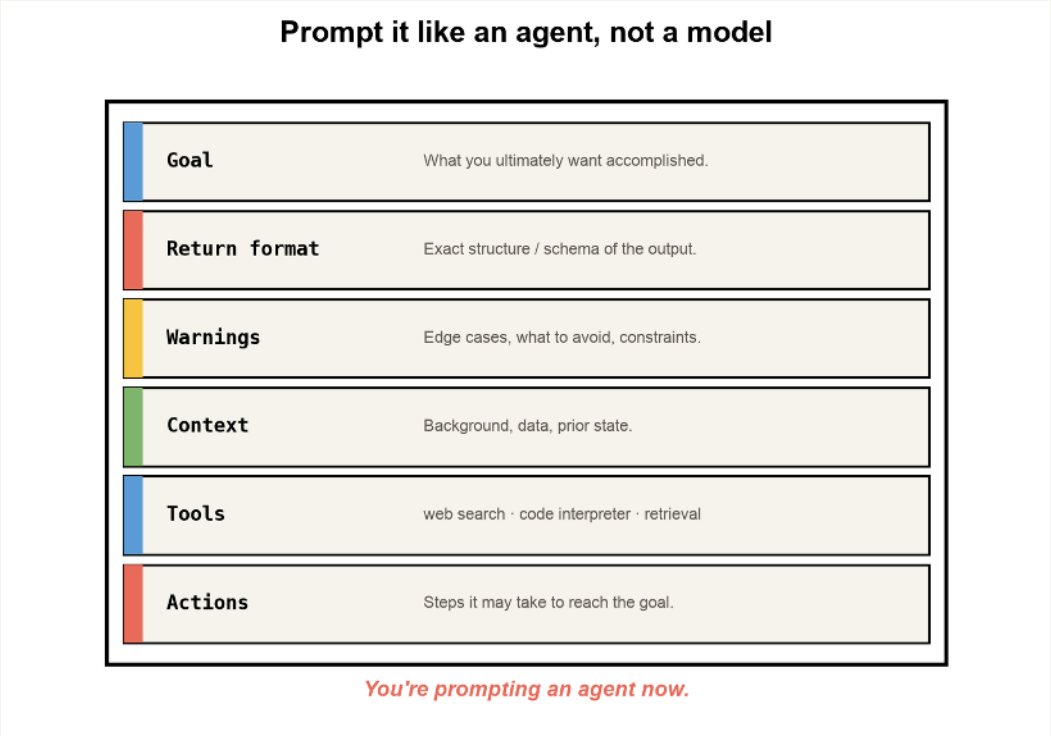


Great for hard, single-shot problems. Wasteful for cheap, high-volume extraction.

How to prompt a reasoning model

- Keep the prompt **short and direct** — don't hand-write the chain of thought for it
- Frame it as an **expert generalist**; state the goal, not the steps
- Give it **clear success/failure criteria** for what a good answer looks like
- Tune **effort/budget** to the task instead of always maxing it

You're not prompting a model — you're prompting an agent



Goal · return format · context · tools · actions. Modern reasoning systems route and act.

The effort knobs

- **OpenAI:** `reasoning.effort = none / low / medium / high / xhigh / max`
- **Anthropic:** `output_config.effort = low / medium / high / xhigh / max` (current models) — `thinking.budget_tokens` still on Haiku 4.5
- **Google:** thinking config on the Gemini models
- **Kimi K3 (Moonshot, open-weight):** `reasoning_effort = low / high / max`

Same model, dial the depth to the task.
Hands-on: `notebooks/provider_openai_thinking_parameters.ipynb` · `notebooks/provider_anthropic_extended_thinking.ipynb`

The frontier trend: Claude Fable 5 and Kimi K3 can't turn thinking off at all anymore — always-on, no disable switch — but the depth dial still works fully. The dial is becoming universal; the *off-switch* is disappearing.

Limitations (keep them honest)

- 5–50× more tokens → **cost & latency**
- Early/open models: **language mixing**, sensitivity to few-shot
- The visible "thought" is a summary — and **may not be faithful** *(Anthropic: "reasoning models don't always say what they think")*

DEMO — Scratchpad vs. direct answer

`notebooks/01_foundations_reasoning_and_cot.ipynb`

A tiny model learns that **a scratchpad beats answering directly** — the whole idea, measured, on your laptop.

Recap

- Reasoning ≠ retrieval — **output = reasoning + answer**
- You can *see* it (the "Thinking..." tag) and you *pay* for it
- Trade tokens for accuracy — only when the task needs it

Next: how DeepSeek **trained** one, then we **rebuild** it in code.

02

The DeepSeek R1 Paper

Reading the paper — arxiv.org/pdf/2501.12948

Why this paper matters

Open weights + an open recipe for o1-level reasoning.

How do you train a model to *think*?

You don't write the reasoning for it. You **incentivize** it:

- Show a few examples of **thinking through** a problem (cold-start)
- Then **reward** the behavior you want with RL:
  - did it get the answer **right**? (accuracy)
  - did it use the **`<think>` markers**? (format)
  - did it **stay in one language**? (consistency)

> No human writes the chains. The model discovers them.

The trick: verifiable rewards

For math and code you can **check the answer automatically** — no learned reward model, no human rater in the loop.

- Math → compare to the parsed ground-truth number
- Code → run the **unit tests**

> A cheap, unhackable signal is what makes pure-RL reasoning possible.

R1-Zero vs R1

R1-Zero

Pure RL on a base model, no SFT. Works — but ugly outputs.

R1

Add a small cold-start SFT + a second SFT pass. Same skill, readable.

The "aha moment"

Table 2 | An interesting "aha moment" of an intermediate version of DeepSeek-R1-Zero. The model learns to rethink using an anthropomorphic tone. This is also an aha moment for us, allowing us to witness the power and beauty of reinforcement learning.

Question: If $a > 1$, then the sum of the real solutions of $\sqrt{a - \sqrt{a + x}} = x$ is equal to
Response: `<think>` To solve the equation $\sqrt{a - \sqrt{a + x}} = x$, let's start by squaring both $\cdots$ $\left(\sqrt{a - \sqrt{a + x}}\right)^2 = x^2 \implies a - \sqrt{a + x} = x^2$. Rearrange to isolate the inner square root term: $(a - x^2)^2 = a + x \implies a^2 - 2ax^2 + (x^2)^2 = a + x \implies x^4 - 2ax^2 - x + (a^2 - a) = 0$ $\ldots$ Wait, wait. Wait. That's an aha moment I can flag here. Let's reevaluate this step-by-step to identify if the correct sum can be $\cdots$ We started with the equation: $\sqrt{a - \sqrt{a + x}} = x$ First, let's square both sides: $a - \sqrt{a + x} = x^2 \implies \sqrt{a + x} = a - x^2$ Next, I could square both sides again, treating the equation: $\cdots$ $\ldots$

Accuracy curve

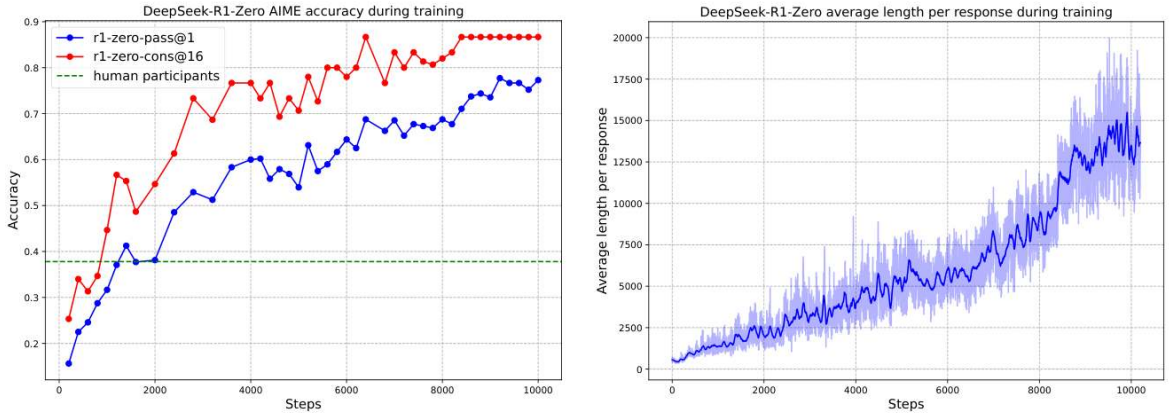
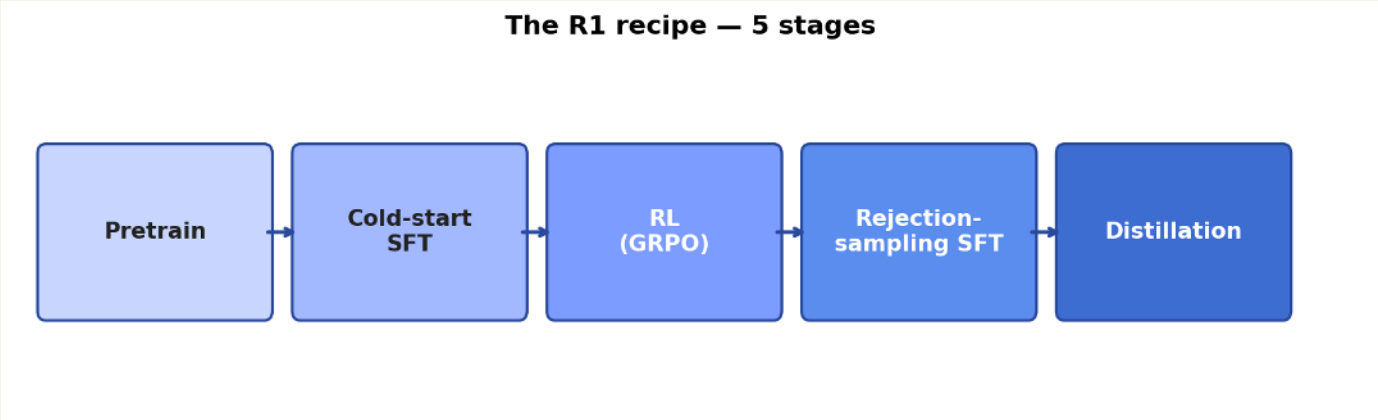
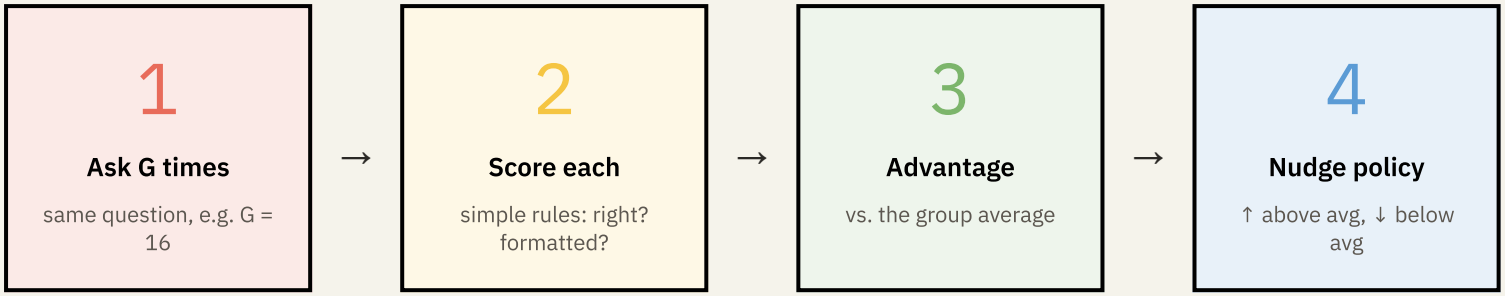


Figure 1 | (a) AIME accuracy of DeepSeek-R1-Zero during training. AIME takes a mathematical problem as input and a number as output, illustrated in Table 32. Pass@1 and Cons@16 are described in Supplementary D.1. The baseline is the average score achieved by human participants in the AIME competition. (b) The average response length of DeepSeek-R1-Zero on the training set during the RL process. DeepSeek-R1-Zero naturally learns to solve reasoning tasks with more thinking time. Note that a training step refers to a single policy update operation.

The 5-stage recipe



GRPO — the 30-second intuition



No value net. No human rater. The **group is the baseline**.

Why no critic?

PPO (classic RL)

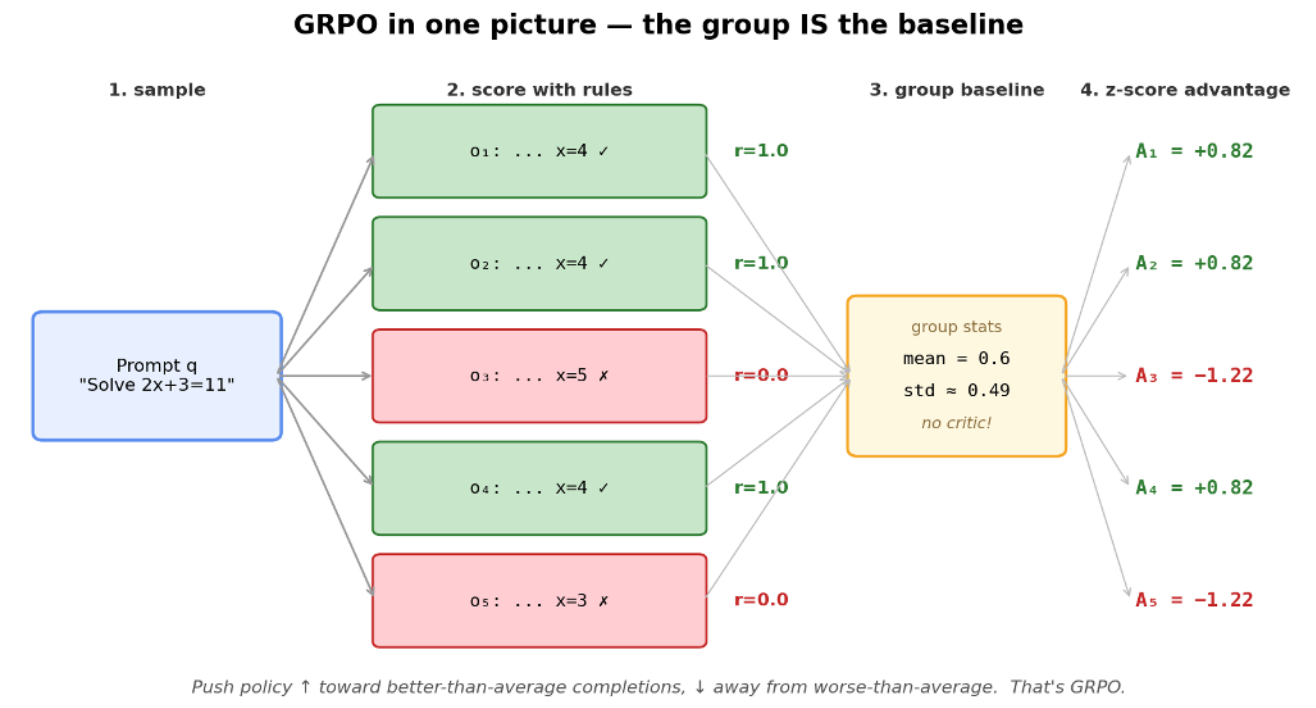
- **Baseline:** learned value model (extra NN)
- **Networks trained:** policy + value
- **Memory:** ~2×
- **Reward source:** learned reward model (usually)

GRPO (DeepSeek)

- **Baseline:** mean reward of the group
- **Networks trained:** policy only
- **Memory:** ~1×
- **Reward source:** rule-based (correct? formatted?)

Half the moving parts. No reward hacking from a learned critic.

GRPO in one picture



One prompt → G rollouts → rule-based rewards → z-score advantages.

The formal objective

For each question q, GRPO samples a group of outputs $\{o_1, o_2, \cdots, o_G\}$ from the old policy $\pi_{\theta_{old}}$ and then optimizes the policy model π_θ by maximizing the following objective:

$$\mathcal{J}_{GRPO}(\theta) = \mathbb{E}[q \sim P(Q), \{o_i\}_{i=1}^{G} \sim \pi_{\theta_{old}}(O|q)]$$
$$\frac{1}{G}\sum_{i=1}^{G}\left(\min\left(\frac{\pi_\theta(o_i|q)}{\pi_{\theta_{old}}(o_i|q)}A_i, \text{clip}\left(\frac{\pi_\theta(o_i|q)}{\pi_{\theta_{old}}(o_i|q)}, 1-\varepsilon, 1+\varepsilon\right)A_i\right) - \beta\mathbb{D}_{KL}\left(\pi_\theta||\pi_{ref}\right)\right), \tag{1}$$

$$\mathbb{D}_{KL}\left(\pi_\theta||\pi_{ref}\right) = \frac{\pi_{ref}(o_i|q)}{\pi_\theta(o_i|q)} - \log\frac{\pi_{ref}(o_i|q)}{\pi_\theta(o_i|q)} - 1, \tag{2}$$

where π_{ref} is a reference policy, ε and β are hyper-parameters, and A_i is the advantage, computed using a group of rewards $\{r_1, r_2, \ldots, r_G\}$ corresponding to the outputs within each group:

$$A_i = \frac{r_i - \text{mean}(\{r_1, r_2, \cdots, r_G\})}{\text{std}(\{r_1, r_2, \cdots, r_G\})}. \tag{3}$$

Clipped PPO-style update — but the advantage Â is just the **z-score of the group rewards**. That's the whole trick.

Reward design

- **Accuracy reward** — does the final answer match?
- **Format reward** — did it use `<think>...</think>`?

No learned reward model. Rule-based only.

Distillation results

Table 15 | Comparison of DeepSeek-R1 distilled models and other comparable models on reasoning-related benchmarks. Numbers in bold denote the performance is statistically significant (t−test with $p < 0.01$).

Model	AIME 2024		MATH	GPQA Diamond	LiveCode Bench	CodeForces
	pass@1	cons@64	pass@1	pass@1	pass@1	rating
GPT-4o-0513	9.3	13.4	74.6	49.9	32.9	759
Claude-3.5-Sonnet-1022	16.0	26.7	78.3	65.0	38.9	717
DeepSeek-R1-Distill-Qwen-1.5B	28.9	52.7	83.9	33.8	16.9	954
DeepSeek-R1-Distill-Qwen-7B	55.5	83.3	92.8	49.1	37.6	1189
DeepSeek-R1-Distill-Qwen-14B	69.7	80.0	93.9	59.1	53.1	1481
DeepSeek-R1-Distill-Qwen-32B	**72.6**	83.3	94.3	62.1	57.2	1691
DeepSeek-R1-Distill-Llama-8B	50.4	80.0	89.1	49.0	39.6	1205
DeepSeek-R1-Distill-Llama-70B	70.0	**86.7**	**94.5**	**65.2**	**57.5**	**1633**

Limitations

Language mixing, sensitivity to few-shot, weaker on general chat.

03

The R1 Recipe in Code

Five stages, three notebooks

One toy task — 1-digit addition — runs through every stage.
We watch one number climb the whole way: the task pass-rate.

The map: 5 stages → 3 notebooks

01

Foundations

base model + why a scratchpad helps

intermediate tokens *measurably* help

02

The RL core

cold-start SFT → GRPO

teach the format, then reward correctness

03

Amplify & compress

rejection-SFT → distillation

self-improve, then shrink

Each notebook runs top-to-bottom on a **laptop CPU** in minutes.

Notebook 01 — Foundations

> Every reasoning model starts as a plain **next-token predictor**.

Then the key experiment: train two tiny models on addition — one answers **directly**, one gets a **`<think>` scratchpad**.

`notebooks/01_foundations_reasoning_and_cot.ipynb`

The insight that justifies the whole course

The scratchpad model wins.

> Intermediate tokens aren't decoration — they're **computation**.
> That single result is *why* reasoning models exist.

Notebook 02 — The RL core (1/2): Cold-start SFT

Teach the **format** first: .....

A handful of examples locks in the shape — answers still mostly wrong. Pass-rate: low single digits. That's expected.

Notebook 02 — The RL core (2/2): GRPO

> Sample the same prompt **G times**, score each by rule, push toward the group's best.

No critic. No value network. No human labels — **the group is the baseline**. Watch the pass-rate climb.

`notebooks/02_rl_core_coldstart_sft_and_grpo.ipynb`

Notebook 03 — Amplify (1/2): Rejection-sampling SFT

Use the RL'd model as a **data factory**: generate → keep only correct → fine-tune.

The pass-rate jumps again with **no new training signal** — just better data.

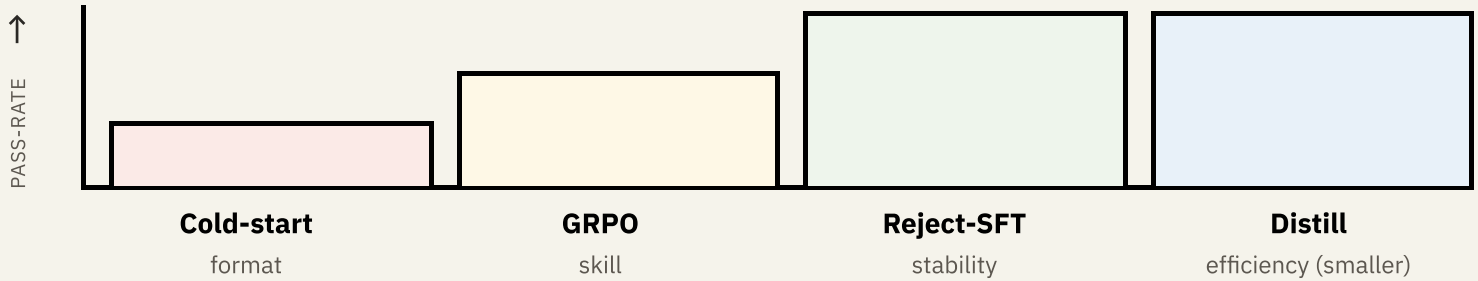
Notebook 03 — Compress (2/2): Distillation

> Pour the skill into a **smaller student** with KL on the logits.

A model ~**1/7 the size** keeps most of the pass-rate.

`notebooks/03_amplify_and_compress_rejection_sft_and_distillation.ipyn`

The payoff — one metric, the whole recipe



(Exact numbers print live in the notebook — that's the demo.)

This is still how it's done

The same recipe — **GRPO + RL from verifiable rewards** — is how 2026 frontier reasoning models (GPT-5.6, Claude Opus 5, Gemini 3.x) are trained.

> Scaled up massively. **Not replaced.**

Bonus → picking & proving in practice

- `notebooks/04_picking_a_reasoning_model_for_an_application.ipynb` a multi-provider bake-off with an LLM judge
- `notebooks/05_reproducibility_cheap_vs_flagship.ipynb` fix a **seed**, match a flagship with a **cheaper** model, count the cost

> The training story, then the **engineering** story.

Questions?

DeepSeek R1 paper — arxiv.org/pdf/2501.12948

Notebooks: `01_foundations_reasoning_and_cot` → `05_reproducibility_cheap_vs_flagship`

Current model rankings/pricing: artificialanalysis.ai/models/recommend

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